BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Unseen Code and Debugging

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the raw mono sound size in bits?

2. What is a character set?

A) A list of fonts
B) Characters linked to codes
C) A list of filenames
D) A collection of pixel colours

3. Given code 65 for A and consecutive capital-letter codes, what is C's code?

4. A website needs English, Greek and Chinese text. "Unicode has fewer characters than standard ASCII." Correct this. Why does Unicode suit this site?

Today's objectives

Explain the core ideas of paper 2: unseen code and debugging using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Read code literally
Do not assume it does what its title claims.

Trace state
Follow nested conditions and loops from initial values.

Identify fault location
Point to the exact condition, bound or assignment.

Repair and retest
State expected outputs for normal and boundary inputs.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Read code literally	Do not assume it does what its title claims.
Trace state	Follow nested conditions and loops from initial values.
Identify fault location	Point to the exact condition, bound or assignment.
Repair and retest	State expected outputs for normal and boundary inputs.

I Do - One worked example

Debug total=1 for a program intended to sum [2,4,6].

1. Trace initial total=1.
2. Add 2,4,6 -> final 13, not 12.
3. Identify unwanted starting value.
4. Set total=0; expected result 12.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A password attempt counter locks after four failures instead of three. Fix it.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the loop bound so the largest file size is found.

```
array files = [6,3,10]
largest = files[0]
for i = 0 to 1
if files[i] > largest then
largest = files[i]
endif
next i
print(largest)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. A loop runs i=0 to 4 inclusive. How many iterations execute? [2 marks]

Q3. Explain why total=1 is wrong for a plain sum accumulator. [2 marks]

Q4. Give a suitable test after fixing an upper boundary error. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Fixing an algorithm should be followed by testing. ☐ T ☐ F
2. Do not assume it does what its title claims. ☐ T ☐ F
3. Follow nested conditions and loops from initial values. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Sample rate x duration in seconds x bits per sample. ☐

Q2 B - Characters linked to codes ☐

Q3 67, because C is two positions after A. ☐

Q4 Unicode supports far more characters than standard ASCII. It can cover English, Greek and Chinese. Use an encoding with enough codes; Unicode characters do not always take one byte. ☐

Code First - Model

```
array files = [6,3,10]
largest = files[0]
for i = 0 to 2
if files[i] > largest then
largest = files[i]
endif
next i
print(largest)
```

Output: 10

We Do - Check the model

1. Test attempts=2 and 3. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 5. ☐

Q3 Adds an unintended 1 to every computed total; initialise to zero. ☐

Q4 Test exactly at upper boundary and one beyond, plus normal value. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: